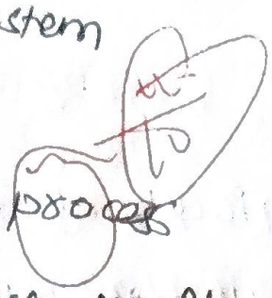
ECA1411 - Embedded system

TUTORIAL-1

TOPIC : Embedded Design process



1) Consider a wearable ECG Monitor with GPRS functionality in a remote healthcare monitoring system. Analyze its design, including the selection of sensors, data processing units, and communication protocols to ensure reliability and accuracy.

Answer :

A wearable ECG Monitor measures the electrical activity of the heart and sends the data to a remote healthcare center.

1. Sensors : ECG electrodes measure the electrical signals of the heath.

2. Processor : A Microcontroller processes the collected ECG signals.

3. Data processing : Noise is removed using digital filters, and heart rate is calculated.

4. Communication : GPRS sends the processed data to a remote server.

5. Power supply : A rechargeable battery powers the device.

6. Reliability : Error detection and secure data transmission help protect patient data.

2. Consider a scenario where an agricultural drone is used to monitor crop health and relay data to a cloud server. Examine the design and operational functionality of its embedded system, including data acquisition, processing, and transmission.